

ALUN REES-LINDLEY, PhD

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Senior Data Scientist & AI Engineer with over 8 years of experience bridging the gap between rigorous statistical methodology and production-ready AI systems. Expertise in Time-Series Methods, Statistics, and AI. Proven track record of leading technical teams within the Civil Service to deliver high-stakes analytical pipelines for national policy. PhD in Computational Physics with a focus on high-performance simulation and numerical modelling.

TECHNICAL EXPERTISE

Statistics & Modeling: Time Series (X13-ARIMA, State-Space Models), Bayesian State-Space Methods, Structural Break Detection, HPC Simulations.

Data Engineering: Python (Expert), R, SQL, Google BigQuery, Azure, Git, Docker, RAP.

AI & Machine Learning: Generative AI (LLMs), Agentic Architectures, LangChain, RAG, XGBoost, TensorFlow/Keras, NLP.

Leadership: Technical Mentorship, Agile/Scrum, Stakeholder Management (Bank of England, Cabinet Office).

PROFESSIONAL EXPERIENCE

Office for National Statistics (ONS) March 2025 – Present
Senior Statistical Methodologist (SEO) Newport & Darlington

- **Engineering Lead:** Architected the migration of legacy economic modeling code to a modular R/GitHub pipeline (RAP) covering ~200 economic series, ensuring 100% auditability for National Accounts (GDP, Retail Sales) and saving ~1 week of manual processing per round.
- **Statistical Innovation:** Developed automated diagnostic software and Bayesian state-space models to handle structural breaks in COVID-impacted tourism data, creating reliable 2-year performance projections.
- **Technical Leadership:** Act as interim G7 Lead; managing work allocation for a team of 3 and providing methodological consultancy to the Bank of England and Treasury analysts.

Government Office for Science October 2022 – March 2025
Senior Data Scientist (SEO) / Data Scientist (HEO) Birmingham & Cardiff

- **GenAI:** Achieved 3rd place in National AI Hackathon; engineered a parallelized, **agentic RAG tool** using LangChain and Claude API to automate legislative drafting.
- **Cloud Engineering:** Re-architected data pipelines by migrating API extractions to **Google BigQuery**, increasing data granularity by 4x and reliability for Prime Minister briefings.
- **NLP & Automation:** Developed spaCy/NLTK pipelines for automated horizon scanning of emerging technologies, supporting the UK’s Horizon Europe membership strategy.
- **Management:** Line-managed HEO staff and oversaw departmental Azure cloud migration projects.

University of Warwick September 2018 – September 2022
PhD Researcher & Computational Assistant Coventry

- Developed high-performance computational features for 2D hydrodynamics code (FORTRAN), managing simulations on HPC clusters to analyze Inertial Confinement Fusion experiments.
- Engineered a sports prediction engine using Neural Networks and Gradient Boosting, achieving ~72% accuracy.

Admiral Group Plc April 2018 – September 2018
Graduate Data Scientist Cardiff

- Developed an imbalanced-class fraud detection model using R and SQL, identifying suspicious claims valued at £10k+ each.
- Reconstructed actuarial pricing models to improve risk-stratification logic and model explainability.

EDUCATION

University of Warwick — PhD in Physics (Fusion Energy - Computational) 2022
University of York — MSc in Fusion Energy (First Class Honours) 2018
Durham University — MPhys in Physics (Upper Second Class Honours) 2016

CERTIFICATIONS, AFFILIATIONS & LANGUAGES

- **Certifications:** AgilePM Foundation (APMG), ONS Data Science Residency.
- **Professional Affiliations:** Royal Statistical Society – Statistics in Sport Committee member; contributor to Significance Magazine.
- **Languages:** English (Native), Welsh (Fluent), French (B1), German (B1).